

Script-Controlled ACL – Restrict Record Access Based on Field Value

3. Project Design Phase

3.1 System Architecture

The design contains four main layers: User Interface, Authentication, ACL Controller, and Database. The user sends a request, authentication identifies the user, the ACL controller evaluates the record fields, and the database operation is performed only when access is allowed.

3.2 Main Modules

- User Authentication Module – verifies the logged-in user.
- User/Role Module – stores user identity, role, department, or branch.
- Record Module – stores application records and their field values.
- ACL Rule Module – contains the access conditions.
- Script Engine/Controller – evaluates the conditions.
- Audit Module – records important access decisions.

3.3 Data Design

A sample Record table can contain Record_ID, Record_Name, Department, Owner_ID, Status, and Confidentiality_Level. A User table can contain User_ID, User_Name, Role, Department, and Branch.

3.4 Access Decision Flow

Request → Authenticate User → Identify Target Record → Read Field Value → Execute ACL Script → ALLOW/DENY → Perform or Block Operation.

3.5 Sample Logic

IF current_user.department = record.department THEN ALLOW ELSE DENY. This logic can be extended with role, ownership, status, or confidentiality conditions.